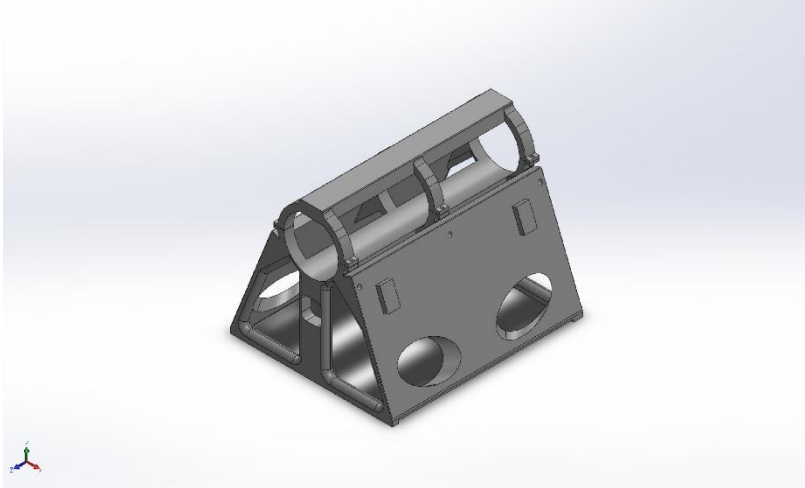




Description.

The goal of the simulation is to model the pressure distribution caused by the incoming current, which has a maximum force per area output of 1027N/m^2 .

From the applied pressure distribution, stress and strain at the hinges and walls have been measured.

Simulation of Chassis Prototype Pressure Distribution

Date: Sunday, February 22, 2026
Designer: Joseph Harb
Study name: Pressure Distribution
Analysis type: Static

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Assumptions

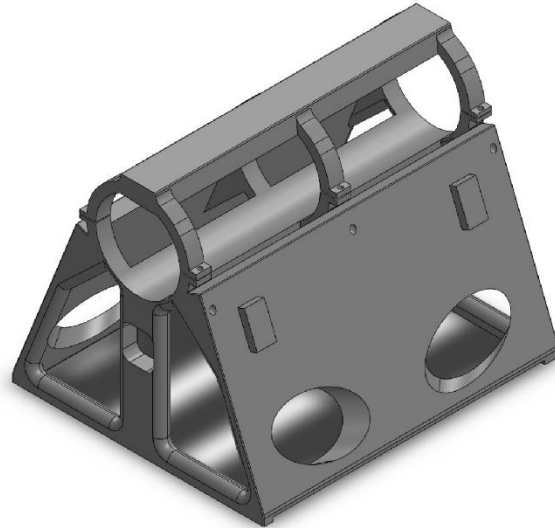
Comments:

Assume that the current acts exactly perpendicular to the surface. This models an overestimation of the stress, and the real effects will not be as exaggerated.

Although the real current varies and is time dependent, this static analysis models the maximum possibly deformity resulting from the current itself.

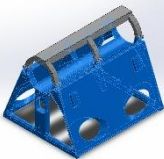
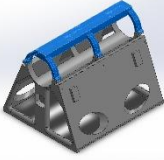


Model Information



Model name: Chassis Prototype (400 mm fixed)
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Force Applied to Bodies 	Solid Body	Mass:3.852 kg Volume:0.003852 m ³ Density:999.999 kg/m ³ Weight:37.7496 N	C:\Users\josep\OneDrive - Texas A&M University\MATE ROV\Final Assembly (20 20 26)\Chassis Prototype (400 mm fixed).SLDPRT Feb 22 00:08:46 2026
Fixture 	Solid Body	Mass:0.588369 kg Volume:0.000588369 m ³ Density:1,000 kg/m ³ Weight:5.76602 N	C:\Users\josep\OneDrive - Texas A&M University\MATE ROV\Final Assembly (20 20 26)\Chassis Prototype (400 mm fixed).SLDPRT Feb 22 00:08:46 2026



Study Properties

Study name	Pressure Distribution
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Contact penalty stiffness scale factor	1
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\josep\OneDrive - Texas A&M University\MATE ROV\Final Assembly (20 20 26))

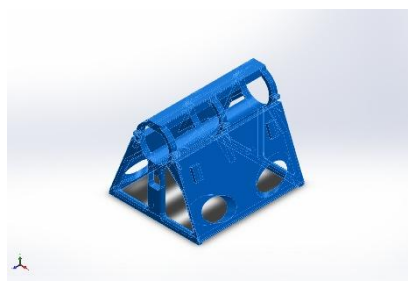


Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²



Material Properties

Model Reference	Properties	Components
	Name: 80% PETG Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 2e+07 N/m ² Tensile strength: 2.3e+07 N/m ² Elastic modulus: 1.7e+09 N/m ² Poisson's ratio: 0.35 Mass density: 1,000 kg/m ³	SolidBody 1(Fillet113)(Chassis Prototype (400 mm fixed)), SolidBody 2(Cut-Extrude47)(Chassis Prototype (400 mm fixed))
Curve Data:N/A		

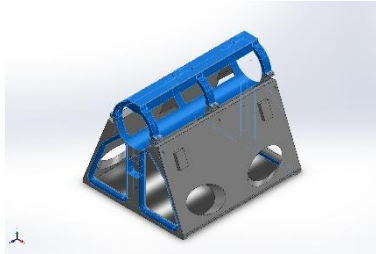
Comments:

PETG 80% Infill,

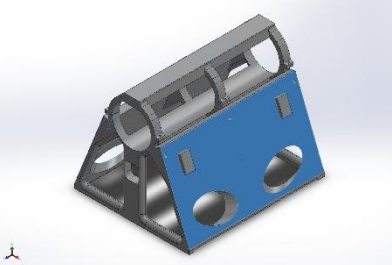
Property	Value	Units
Elastic Modulus	1700000000	N/m ²
Poisson's Ratio	0.35	N/A
Shear Modulus		N/m ²
Mass Density	1000	kg/m ³
Tensile Strength	23000000	N/m ²
Compressive Strength		N/m ²
Yield Strength	20000000	N/m ²
Thermal Expansion Coefficient		/K
Thermal Conductivity	0.2256	W/(m·K)
Specific Heat	1386	J/(kg·K)



Loads and Fixtures

Main Fixtures	Fixture Image	Fixture Details
Electronics Enclosure Fixed		Entities: 4 edge(s), 77 face(s) Type: Fixed Geometry

Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-0.00400313	56.9111	-0.018106	56.9111
Reaction Moment(N.m)	0	0	0	0

Side Walls	Load Image	Load Details
Pressure on Both Faces		Entities: 2 face(s) Type: Normal to selected face Value: 1,027 Units: N/m ² Phase Angle: 0 Units: deg

Connector Definitions

No Data



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	30.4222 mm
Minimum element size	1.52111 mm
Mesh Quality	High

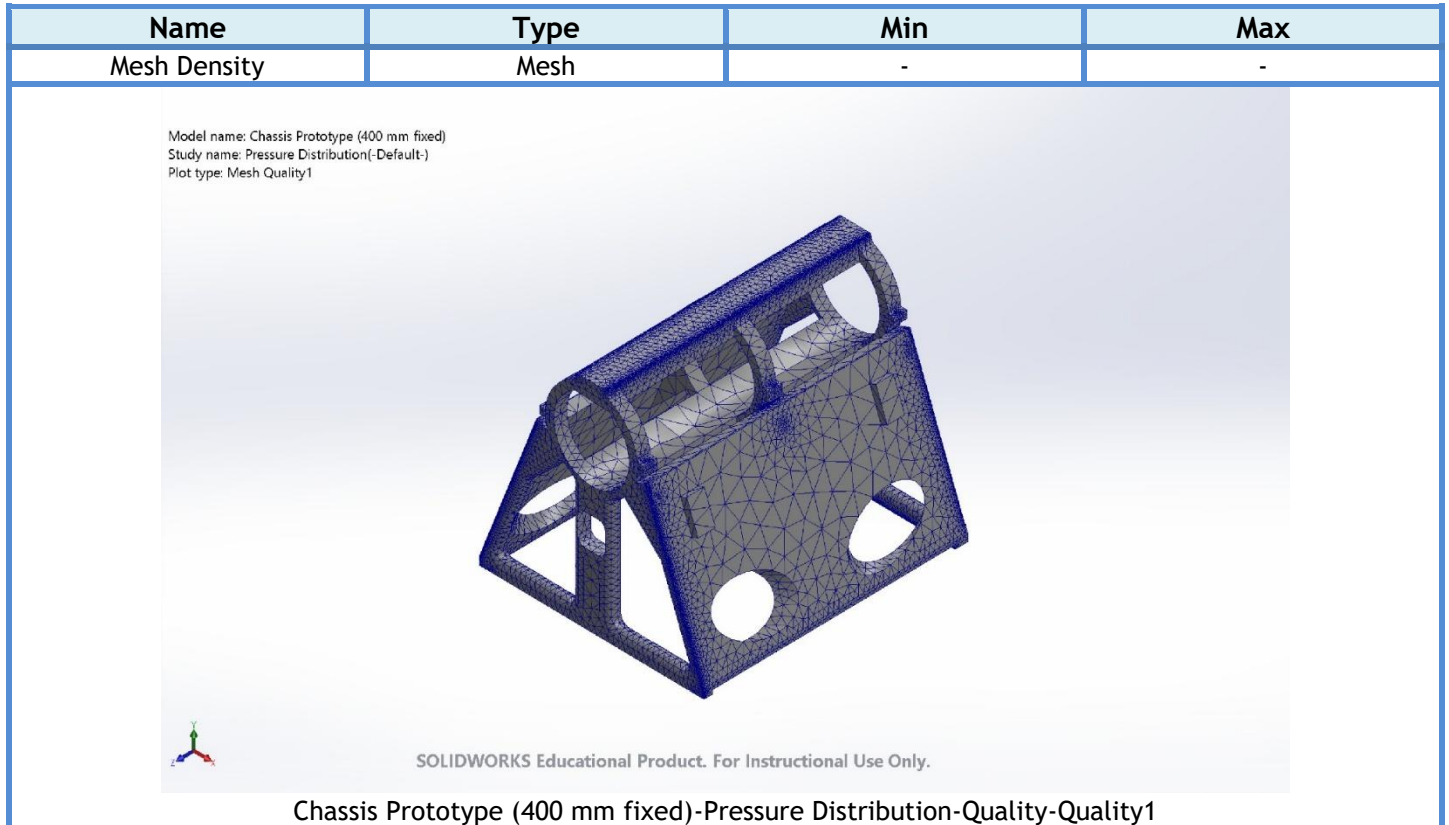
Mesh information - Details

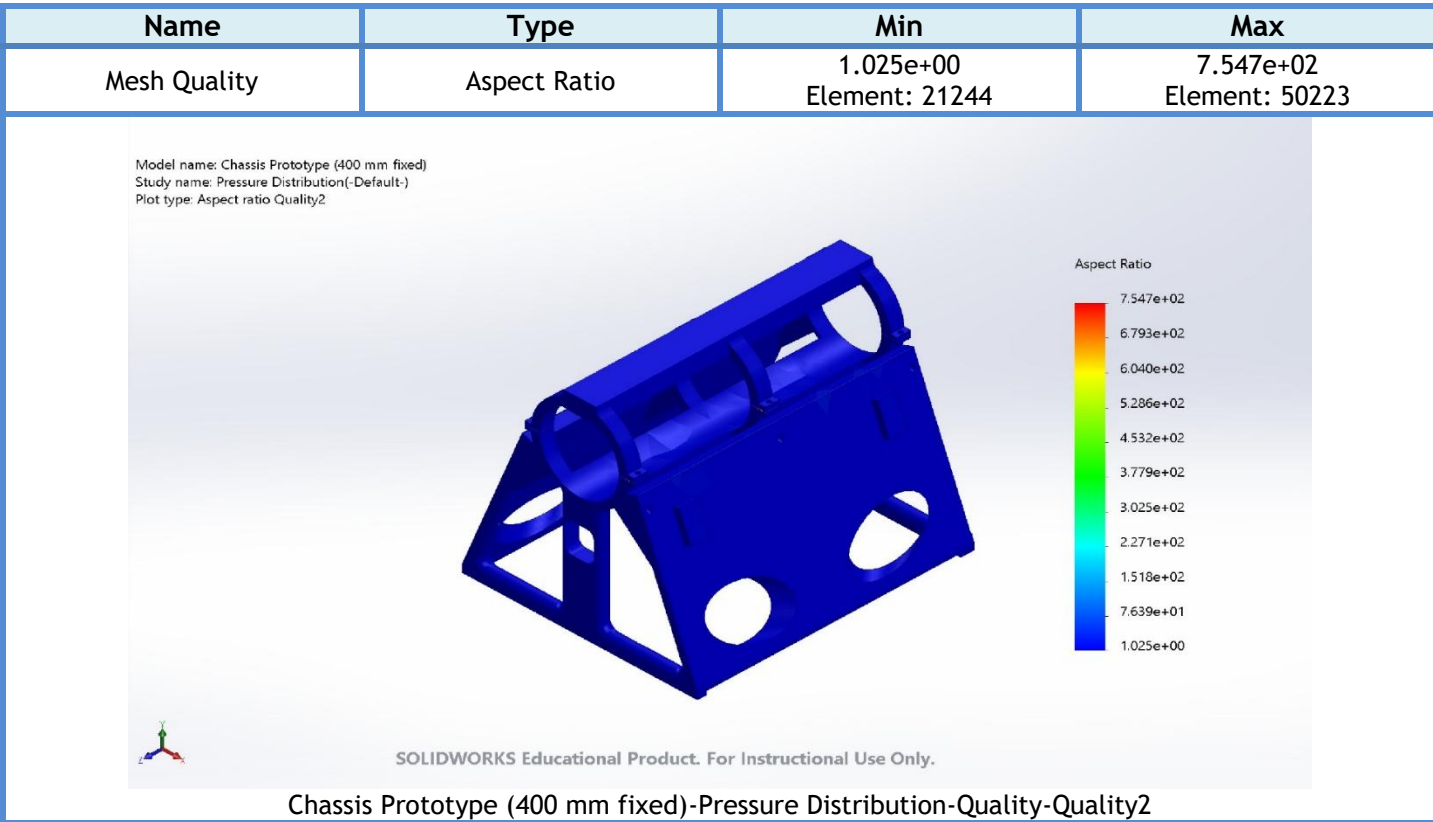
Total Nodes	160252
Total Elements	93827
Maximum Aspect Ratio	754.7
% of elements with Aspect Ratio < 3	96.8
Percentage of elements with Aspect Ratio > 10	0.439
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:08
Computer name:	JOSEPHHARB



Mesh Quality Plots

Comments: The mesh quality is rated high, corresponding to an accurate simulation. The highest aspect ratios are located within the electronics enclosure, which has been modeled as a fixture, and does not interfere with the results of the simulation itself.





Sensor Details

Mesh Details:

Study name: Pressure Distribution (-Default-)

Mesh type: Solid Mesh

Mesher Used: Blended curvature-based mesh

Jacobian points for High quality mesh: 16 points

Max Element Size: 30.4222 mm

Min Element Size: 1.52111 mm

Mesh quality: High

Total nodes: 160252

Total elements: 93827

Maximum Aspect Ratio : 754.7

Percentage of elements\with Aspect Ratio < 3 : 96.8

Percentage of elements\with Aspect Ratio > 10 : 0.439

Percentage of distorted elements : 0

Number of distorted elements:0

Reuse mesh for identical bodies: Off

Number of bodies that have reused mesh:0

Time to complete mesh(hh:mm:ss): 00:00:08



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.00400313	56.9111	-0.018106	56.9111

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.241517	-0.0393057	0.0940476	0.262145

Free body moments

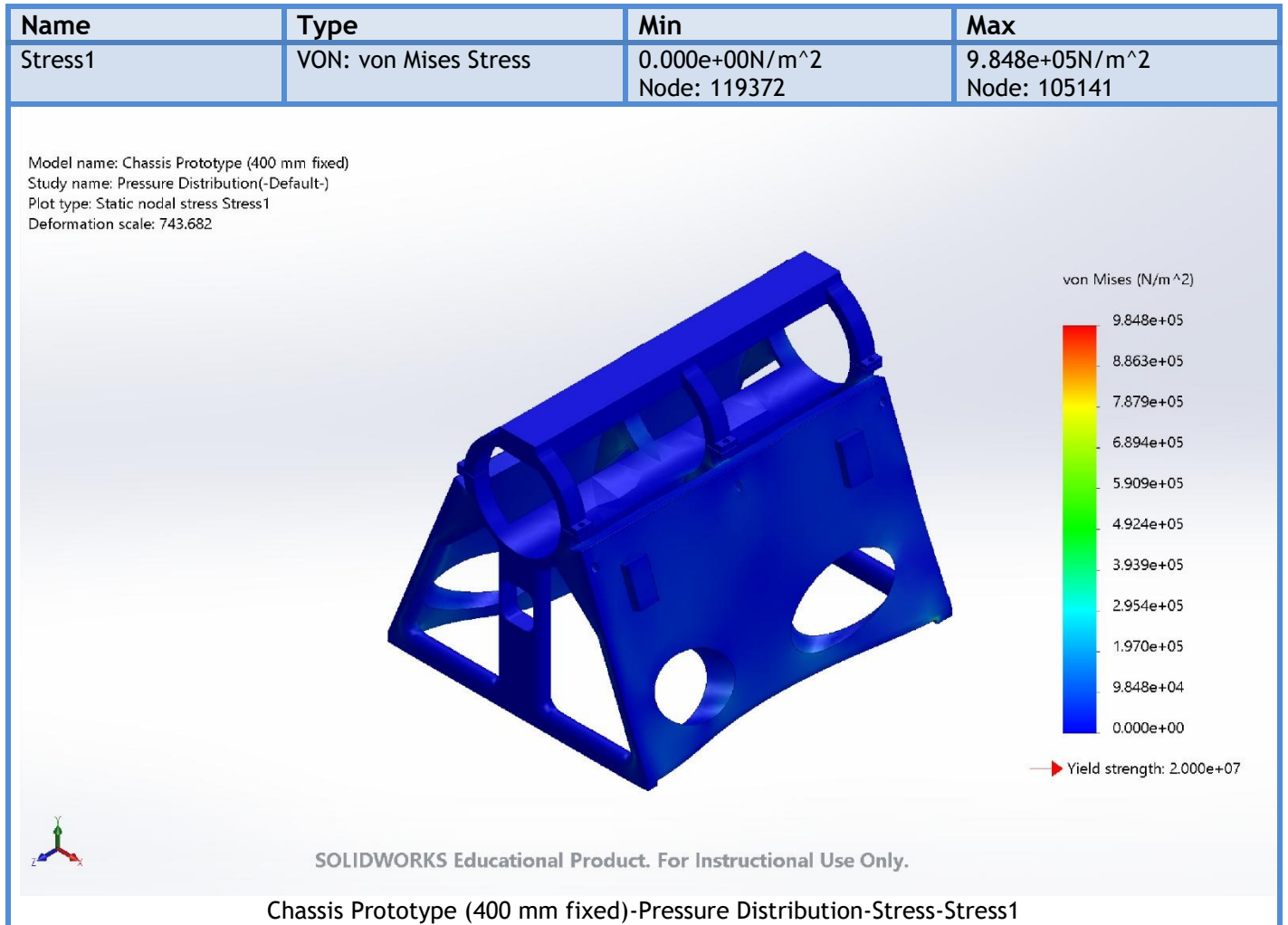
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

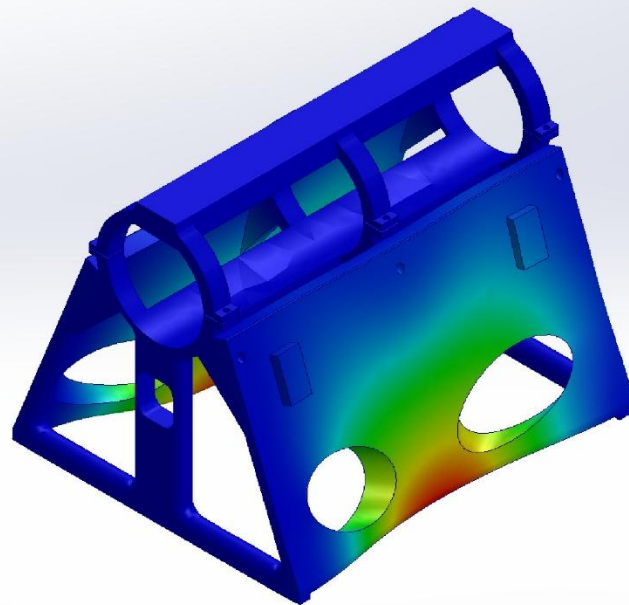


Study Results

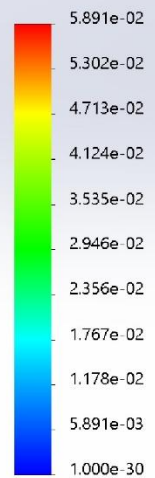


Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 10	5.891e-02mm Node: 116058

Model name: Chassis Prototype (400 mm fixed)
Study name: Pressure Distribution(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 743.682



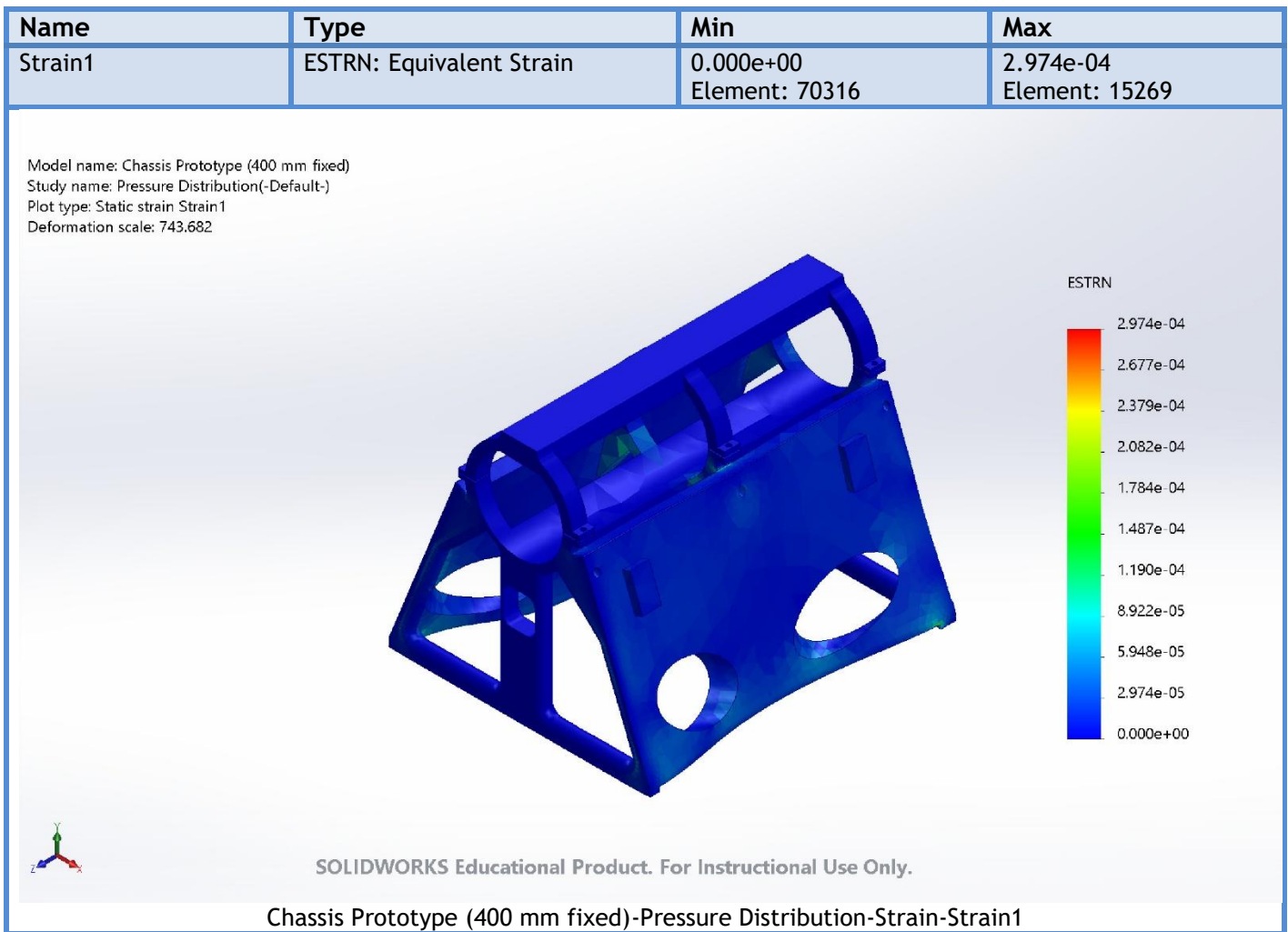
URES (mm)



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Chassis Prototype (400 mm fixed)-Pressure Distribution-Displacement-Displacement1





Conclusion

The deformation of the body is extremely exaggerated to emphasize effects.

The maximum displacement of the entire body was **0.05891 mm**, which is negligible for our purposes.

The main thing to take away from this simulation is that **the geometry connecting the electronics enclosure to the chassis will not deform or break from external pressure to the walls of the chassis.**

The thickness of the walls themselves are adequate and will not deform against the provided current.

